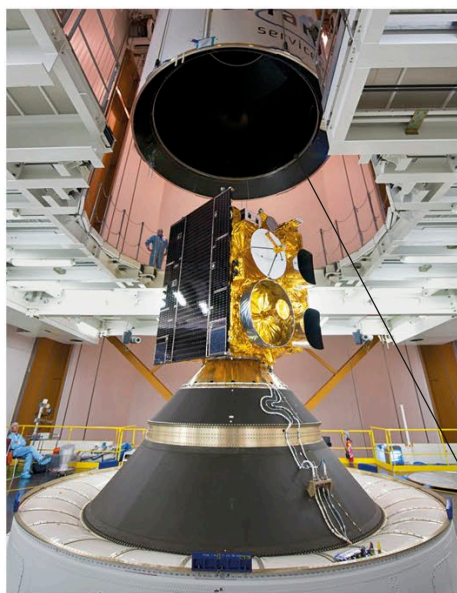


# Satellites

A satellite is a natural or artificial object that orbits another. Just as the moon orbits Earth, so do man-made satellites that are launched into space on a trajectory where they move at high speed but are held on more or less circular paths by the tug of Earth's gravity. Since the 1950s, satellites have revolutionized many aspects of our everyday lives and improved our knowledge of Earth and the wider universe.



Loading the Insat 3D satellite, 2013

## REACHING ORBIT

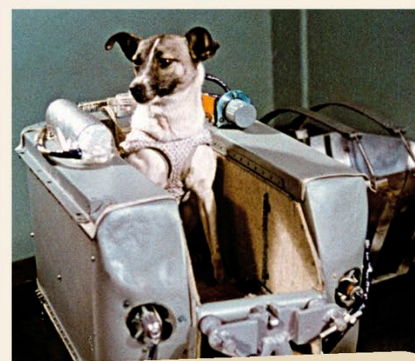
Putting a satellite into orbit required new, more powerful rockets, such as the R-7 missile. Today, satellites such as the Insat 3 weather satellite are launched by towering multistage rockets. After the lower stages of the rocket have placed the satellite in an initial orbit, an upper stage motor fires to boost it to its final trajectory.

*Streamlined cover called a "fairing" is lowered over satellite prior to launch*



## FIRST SATELLITE

Although some earlier rockets had briefly reached space before falling back to Earth, the first object to stay in orbit was Sputnik 1, launched by Russia in October 1957. A simple, football-sized sphere equipped with antennae and a battery-powered radio transmitter, it heralded the start of the space age.



## ANIMALS IN ORBIT

Some early satellites carried animals on board to test how spaceflight might affect humans. Launched in 1957, Sputnik 2 carried a small dog called Laika. She was sent into space with no hope of return, but most later animal astronauts wore specially made spacesuits and returned safely—including two chimpanzees sent into space by NASA in 1961.